EC-110: COMPUTING FUNDAMENTALS

MEMORY AND STORAGE DEVICES

MEMORY & STORAGE DEVICES

- Memory
- Storage
- Primary Devices
- Secondary Devices
- ROM (Read Only Memory)
- RAM (Random Access Memory)

Memory

The term memory identifies data storage that comes in the form of chips

Storage

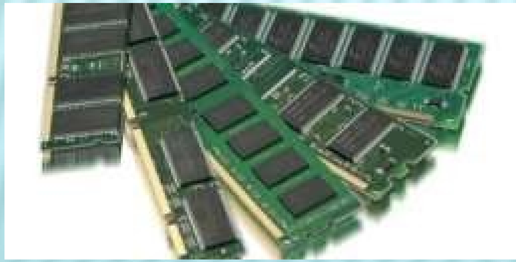
The word storage is used for memory that exists on tapes or disks. Moreover, the term memory is usually used as a shorthand for physical memory, which refers to the actual chips capable of holding data.

MISCONCEPTION

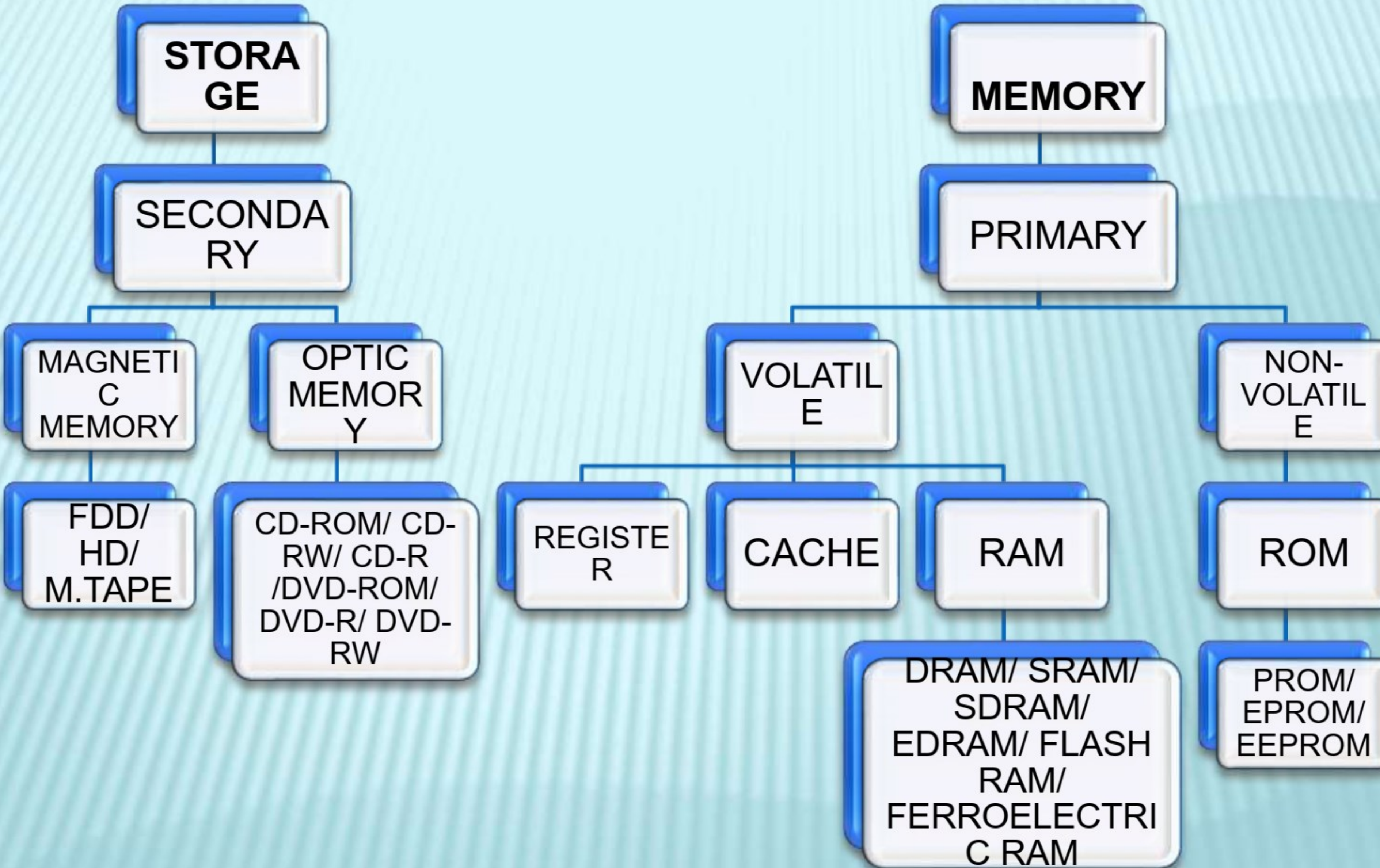
Understandably, many computer users consider memory and storage to be the same thing. Those who realize that there is a difference between the two often cannot identify this difference. If you are unsure about the difference between memory and storage in computers, take a look to the next slide.

Memory	Storage
The term memory refers to the component within your computer that allows you to access data that is stored for a short term.	Storage is the component of your computer that allows you to store and access data on long term basis.
Your computer performs many operations by accessing data stored in its memory. E.g editing a document, loading application and internet browsing etc.	Storage allows you to access and store your applications, operating system and files for an indefinite period of time.
You may recognize this component as DRAM.	Storage comes in the form of a solid-state drive or a hard drive.
The numbers of chips represents the computer's memory.	The filling cabinet represents the storage of your computer.

Memory	Storage
The speed and performance of your system depends on the amount of memory that is installed on your pc.	Items that must be kept yet won't necessarily be accessed soon are stored in the filling cabinet. Due to the size of the filling cabinet, many things can be stored.
The former clears when the computer is turned off. Any files that are left on your desk when you leave the office will be thrown away.	Everything in your filling cabinet will remain. No matter how many times you turn off your computer.



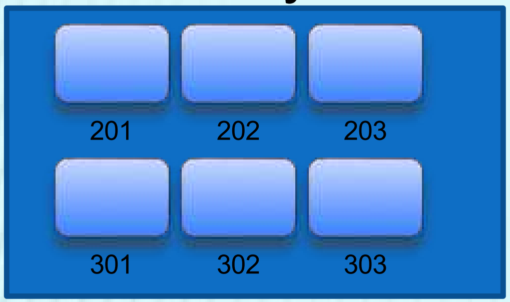
Memory and Storage are not the same (Proved)

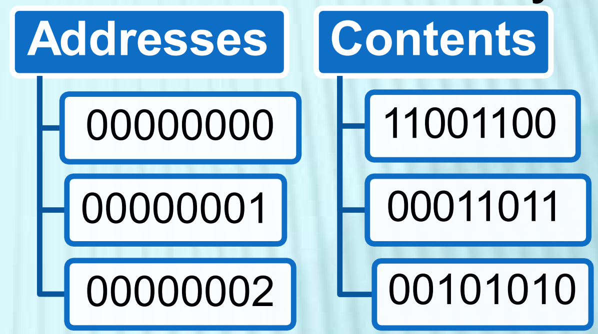


STORAGE
SECONDARY
MAGNETIC MEMORY
OPTIC MEMORY
FDD/ HD/ M.TAPE
CD-ROM/ CD-RW/ CD-R /DVD-ROM/ DVD-R/ DVD-RW
MEMORY
PRIMARY
VOLATILE
NON-VOLATILE
REGISTER
CACHE
RAM
ROM
DRAM/ SRAM/ SDRAM/ EDRAM/ FLASH RAM/ FERROELECTRIC RAM
PROM/ EPROM/ EEPROM

Memory Addresses:

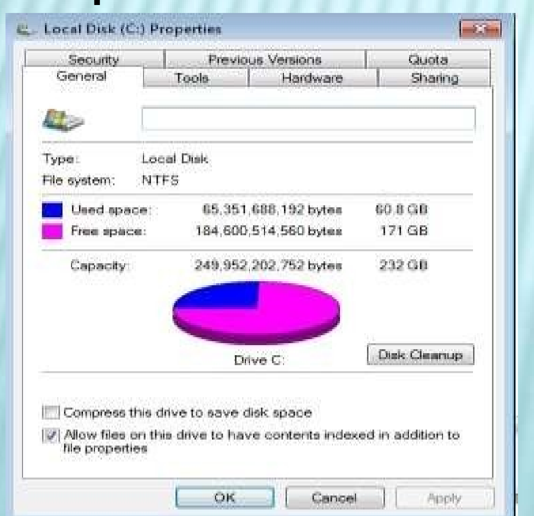
Memory is a collection of cells, each with a unique physical/memory address. Each cell can hold one byte or 8 bits.





Storage Capacity:

The amount of data a storage device such as a disk or tape can hold.



Prefixes for bit and byte multiples

Decimal		Binary		
Value	SI	Value	IEC	JEDEC
1000	k kilo-	1024	Ki kibi-	K kilo-
1000^2	M mega-	1024^2	Mi mebi-	M mega-
1000^3	G giga-	1024^3	Gi gibi-	G giga-
1000^4	T tera-	1024^4	Ti tebi-	
1000^5	P peta-	1024^5	Pi pebi-	
1000^6	E exa-	1024^6	Ei exbi-	
1000^7	Z zetta-	1024^7	Zi zebi-	
1000^8	Y yotta-	1024^8	Yi yobi-	

1 KB	1024 B
1 MB	1024 KB
1 GB	1024 MB
1 TB	1024 GB
1 PB	1024 TB
1 EB	1024 PB
1 ZB	1024 EB
1 YB	1024 ZB

B = byte
KB = Kilobyte
MB = Megabyte
GB = Gigabyte
TB = Terabyte
PB = Petabyte
EB = Exabyte
ZB = Zettabyte
YB = Yottabyte

PRIMARY (SEMI-CONDUCTOR CHIP) MAIN MEMORY

Volatile Memory (temporary):

Volatile memory *requires electricity or some kind of current* to store information. Most common example of Primary *storage volatile memory is RAM*. With RAM, the operating system only uses this memory during operations. *Once the power is turned off, the RAM is wiped clear*. (Cache and Registers are also volatile memory.)

Non-Volatile Memory (permanent):

Non-Volatile memory does not require any kind of power to hold information . Most common example of Non-*volatile memory is ROM*. It is used to *store data and information for permanent form. No matter how many times the computer is turned Off.*

SECONDARY DEVICES

This is also called *Mass Storage, Auxiliary Memory* and *External Memory*. This memory is *slower than the Main memory* as it involves mechanical motion techniques during storage and retrieval of data. This memory is *larger in size than Main memory*. Data from secondary storage must be loaded into RAM before the processor starts processing it. The main memory links the secondary memory to the processor.

Secondary devices can be further categorize in:

- *Magnetic Storage Memory*
- *Optical Storage Memory*

MAGNETIC STORAGE DEVICES & THEIR WORK

Magnetic Storage is one of the most affordable way to store large amounts of data and has been implemented using magnetic tape, floppy disks & hard disk. *Magnetic storage devices hold data, even when the computer is turned off*. Once the grains have been aligned by an external magnetic field, the information remains stored for long periods of time.

There are *three major types of Magnetic Storage* Devices given below:

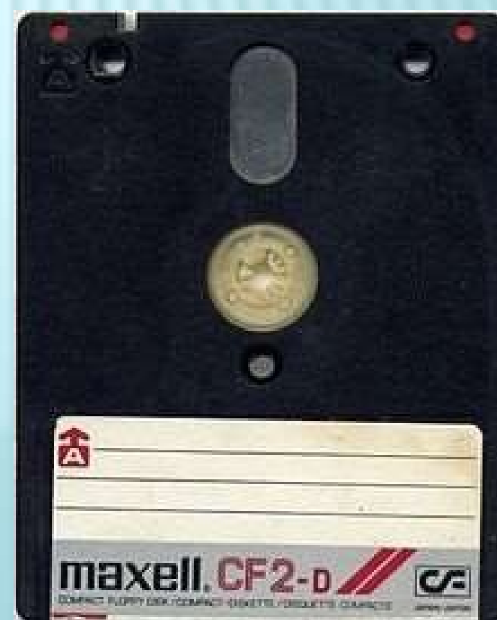
- Magnetic Tape Drive
- Floppy Disk Drive (FDD)
- Hard Disk Drive (HDD)

MAGNETIC TAPE DRIVE

A *magnetic tape drive is a storage device that makes use of magnetic tape as a medium for storage*. It uses a long strip narrow plastic film with tapes of thin magnetisable coating*. It is essentially a device which records or perhaps plays back video and audio using magnetic tape*.

FLOPPY DISK DRIVE

Floppy Disk is *also called Diskette drivers*, *read and write to diskettes* (*called floppy disks or floppies*).In disks the areas to save *data are organized as a set of concentric circles called Tracks*.



HARD DISK DRIVE

Hard Disk composed of one or more platters that are permanently sealed within a hard metallic casing. These *hard disks are fixed in the computer CPU (Nowadays, Also Available with USB port)* and are seldom transferred from one computer to another. *For the better use of the hard disk space, a hard disk can be divided into any number of partitions like C: D: E: etc*.

OPTICAL STORAGE

An *optical disk is a high-capacity storage medium*. An optical drive uses reflected light to read data. To store data, the disk's metal surface is covered with tiny dents (pits) and flat spots (lands), which cause light to be reflected differently. *Optical storage media includes CD-ROM, DVD-ROM, DVD-RAM etc.*

CD-R and CD-RW Discs and Recorders

CD-R

- Stands for Compact Disc- Recordable.
- Discs can be read and written to.
- Discs can only be written to "once".
- CD-R drives are capable of reading and writing data.

CD-RW

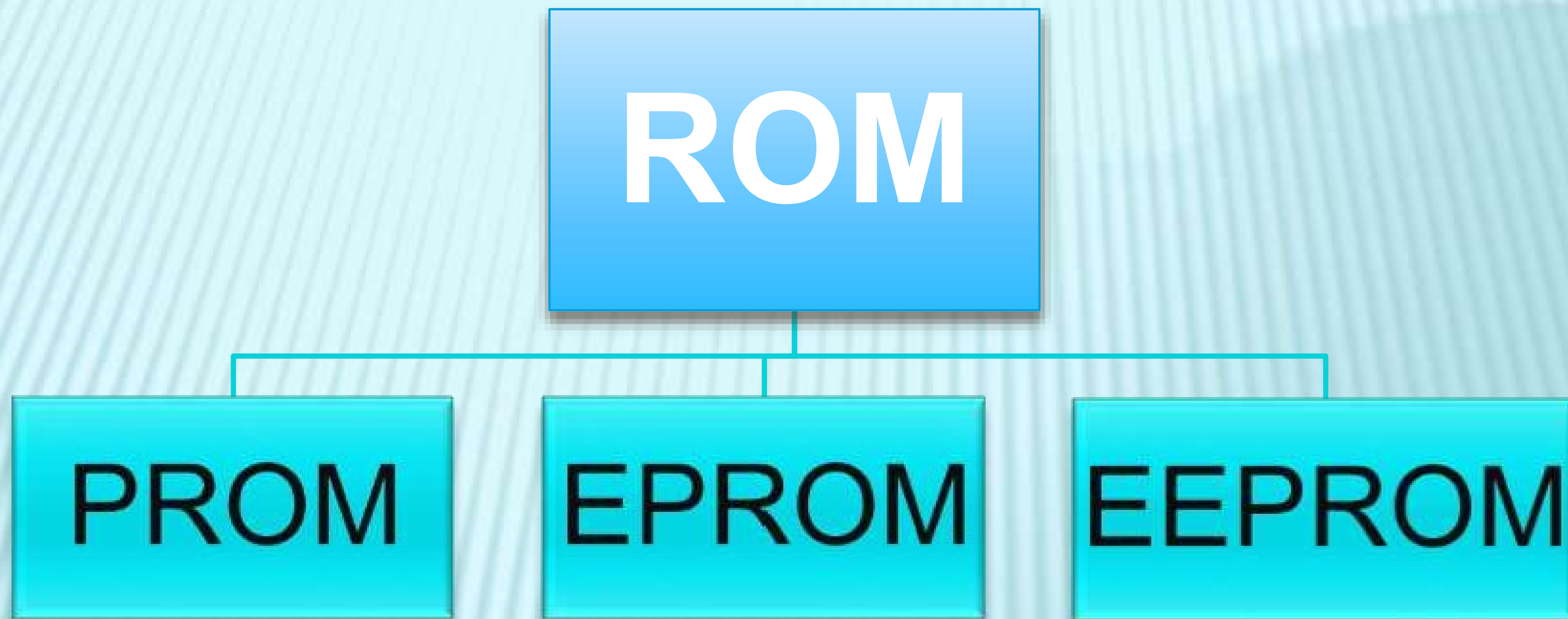
- Stands for Compact Disc- Re-Writeable.
- Discs can be read and written to.
- Discs are erasable.
- Discs can be written to many times.
- CD-RW drives are capable of reading, writing, and erasing data.

DVD-ROM Discs and Drives

- DVD stands for Digital Video Disc.
- DVD technology is similar to CD-ROM technology.
- DVDs are capable of storing up to 17GB of data.
- The data transfer rate of DVD drives is comparable to that of hard disk drives.
- DVD-R and DVD-RW drives have the ability to read/write data.

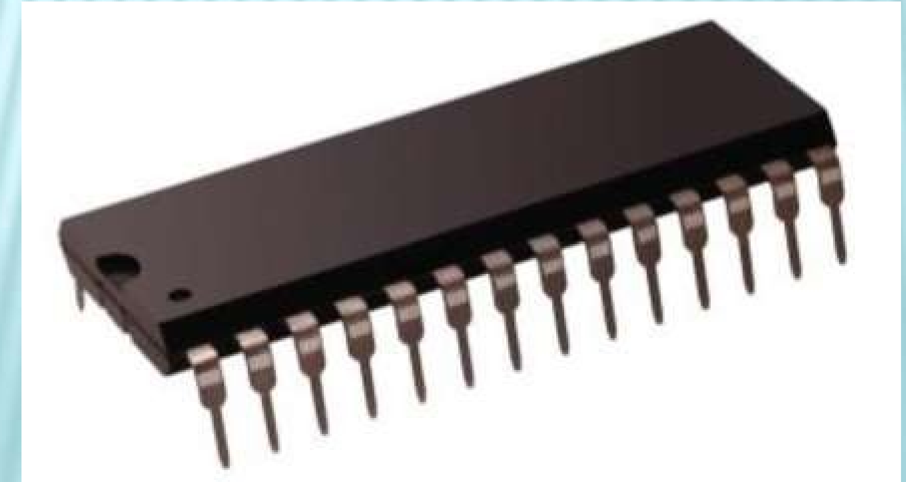
WHAT IS ROM ?

Read Only Memory *ROM is nonvolatile memory*. The contents in locations in ROM cannot be changed. *the data in ROM is not lost when the computer power is turned off*. The *ROM is sustained by a small long-life battery in your computer.*


ROM
PROM
EPROM
EEPROM

PROM- PROGRAMMABLE READ-ONLY MEMORY

Programmable read-only memory (PROM) is read-only memory (ROM) that *can be modified once by a user*. PROM is a way of *allowing a user to tailor a microcode program using a special machine called a PROM programmer* . This machine supplies an electrical current to specific cells in the ROM that effectively blows a fuse in them. The process is known as burning the PROM.



EPROM- ERASABLE PROGRAMMABLE ROM

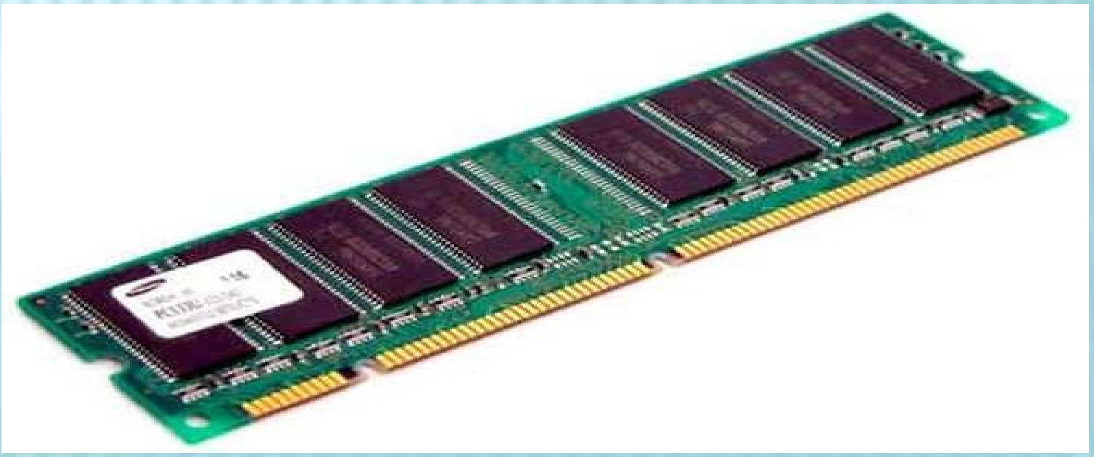
EPROM (erasable programmable read-only memory) is programmable read-only memory (programmable ROM) that *can be erased and re-used*. Erasure is caused by shining an intense ultraviolet light through a window that is designed into the memory chip.

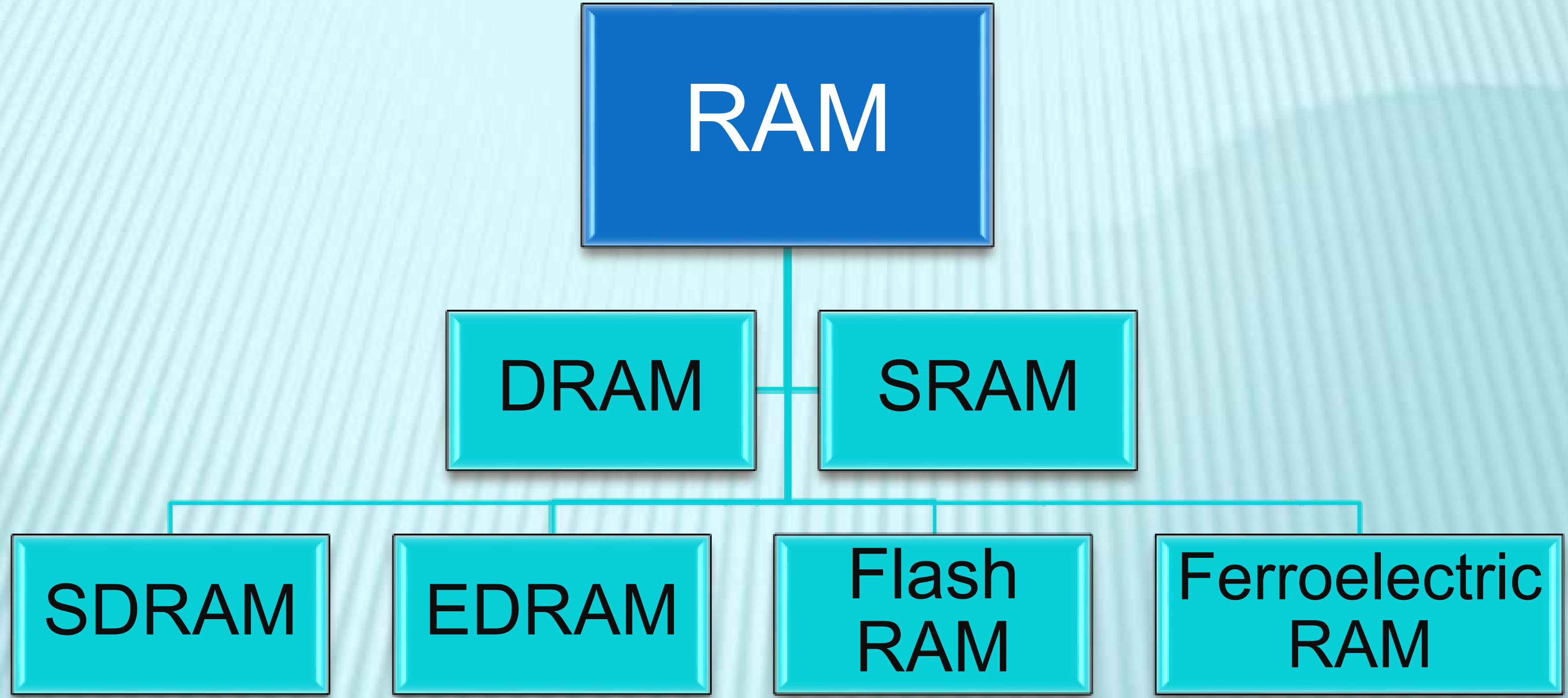
EEPROM- ELECTRICALLY ERASABLE PROM

EEPROM (Electrically Erasable PROM) is user-modifiable ROM that can be erased and reprogrammed (written to) repeatedly through the application of higher than normal electrical voltage. Through *EEPROM, an individual byte of data can erase and reprogrammed entirety, not selectively by the electrical voltage*. For modification in the EEPROM chip, there is no need of removing the chip from the computer. *EEPROM perform read and write cycle very slowly as compared to the read and write cycles of RAM. Here, erase and write operations are performed by byte per byte.*

WHAT IS RAM ?

Random Access Memory or *RAM is a type of volatile memory*. *which stores frequently used program instructions to increase the general speed of a system*. *Data in RAM is not permanently written. When you power off your computer the data stored in RAM is deleted*. RAM is a form of data storage that can be accessed randomly at any time, in any order and from any physical location.




RAM
DRAM
SRAM
SDRAM
EDRAM
Flash RAM
Ferroelectric RAM

DRAM-DYNAMIC RAM

DRAM is a type of memory that is typically used for data or program code that a computer processor needs to function. *DRAM is a common type of random access memory (RAM) used in personal computers, workstations and servers. It uses separate capacitor to store each bit of data. It is used in Main memory.*

SRAM-STATIC RAM

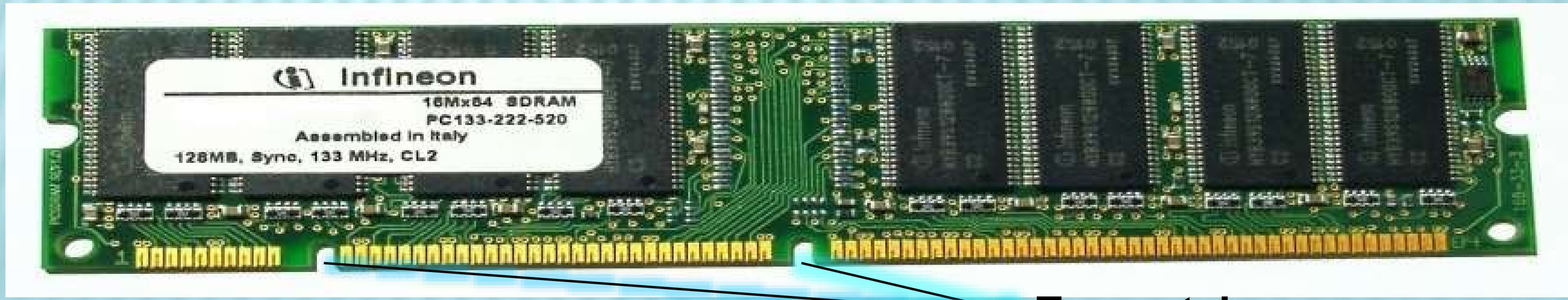
SRAM is random access memory that retains data bits in its memory as long as power is being supplied. *Unlike dynamic RAM (DRAM), which stores bits in cells consisting of a capacitor and a transistor, SRAM does not have to be periodically refreshed*. *Static RAM provides faster access to data and is more expensive than DRAM.*

Parameter	SRAM	DRAM
Full Form	Static Random Access Memory	Dynamic Random Access Memory
Read & Write speed	Faster	Slower than SRAM
Storage component	Uses transistor to store single bit of information	Uses separate capacitor to store each bit of data
Price	Expensive than DRAM	Economical than SRAM
Power consumption	More	Less
Refresh	No need to refresh for maintaining data	Needs to be refreshed thousands of time every second
Used in	Cache memory	Main memory
Internal structure	Complex	Simpler than SRAM
Density	Less dense	Highly dense
Storage per bit	Can store many bits per chip	Cannot store many bits per chip

SDRAM-SYNCHRONOUS DYNAMIC RAM

SDRAM is a generic name for various kinds of dynamic random access memory (DRAM) that are synchronized with the clock speed that the microprocessor is optimized for. This tends to increase the number of instructions that the processor can perform in a given time. The speed of SDRAM is rated in MHz rather than in nanoseconds (ns). This makes it easier to compare the bus speed and the RAM chip speed.

You can convert the RAM clock speed to nanoseconds by dividing the chip speed into 1 billion ns (which is one second). For example, an 83 MHz RAM would be equivalent to 12 ns.



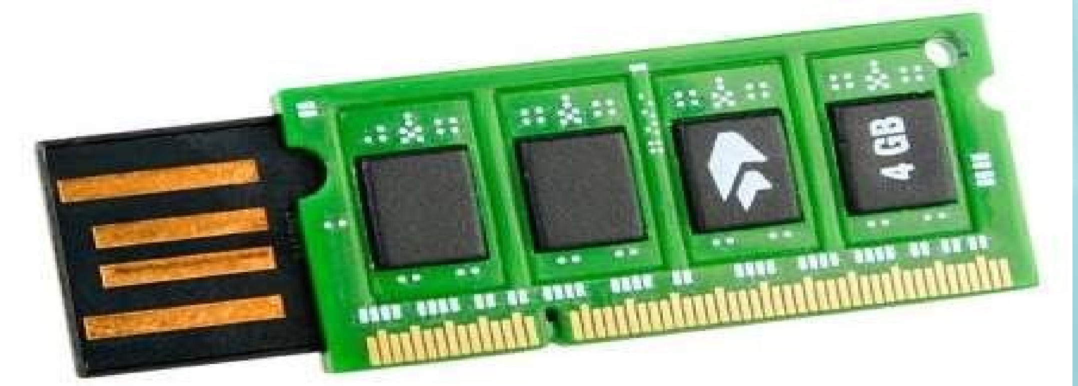
Two notches

EDRAM- ENHANCED DYNAMIC RAM

EDRAM is dynamic random access memory that includes a small amount of static RAM (SRAM) inside a larger amount of DRAM so that many memory accesses will be to the faster SRAM. EDRAM is sometimes used as L1 and L2 memory and, together with Enhanced Synchronous Dynamic DRAM

FLASH RAM

A special type of memory that works like both RAM and ROM. You can write information to flash memory, like you can with RAM, but that information isn't erased when the power is off, like it is with ROM.



FERROELECTRIC RAM

FRAM is random access memory that combines the fast read and write access of dynamic RAM (DRAM) - the most common kind of personal computer memory with the ability to retain data when power is turned off (as do other non- volatile memory devices such as ROM and flash memory). Because FRAM can not store as much data in the same space as DRAM and SRAM, it is not likely replace these technologies. However, because it is fast memory with a very low power requirement

DIFFERENCE BETWEEN RAM & ROM

RAM	ROM
RAM is Random Access Memory.	ROM IS READ ONLY MEMORY
RAM is the memory available for the operating system, programs and processes to use when the computer is running.	ROM is the memory that comes with your computer that is pre-written to hold the instructions for booting-up the computer.
RAM requires a flow of electricity to retain data (e.g. the computer powered on).	ROM will retain data without the flow of electricity (e.g. when computer is powered off).
RAM is a type of volatile memory. Data in RAM is not permanently written. When you power off your computer the data stored in RAM is deleted.	ROM is a type of non- volatile memory. Data in ROM is permanently written and is not erased when you power off your computer.

RAM	ROM
Any memory location can be accessed in a random way	Any memory location can not be accessed in a random way
RAM is analogous to a blackboard on which information can be written with a chalk and erased any number of times.	While ROM is permanent and can only be read.
RAM allows the computer to read data quickly to run applications. It allows reading and writing.	ROM stores the program required to initially boot the computer. It only allows reading.
Physically the size of RAM chip is larger then ROM.	Physically the size of ROM chip is smaller then RAM.

Thank you